

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 1.1: What is AI?

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Artificial Intelligence is ...

*"... technologies that complement and **strengthen human capabilities** in seeing, hearing, analyzing, deciding, and acting."*

Translated from Microsoft, [Was ist künstliche Intelligenz?](#)
Definition & Funktionen von KI

"... a poor choice of words in 1954."

Someone on Twitter (now "X") cited by [Sci-fi writer Ted Chiang](#)

*"... a **field of research in computer science** that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that **maximize their chances** of achieving defined goals."*

Russel & Norvig, [Artificial Intelligence: A Modern Approach](#)

For whom is AI used and with what goal?

**ethical aspects
of AI**

*"... the ability of a machine to **display human-like capabilities** such as reasoning, learning, planning and creativity."*

European Parliament, [What is artificial intelligence and how is it used?](#)

Artificial Intelligence is ...

methods & software

- code
- algorithms
- programs

maximize chances

- probability
- optimization

defined goals

- grasp object
- recognize number
- generate text
- maintain engagement

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Russel & Norvig, [Artificial Intelligence: A Modern Approach](#)

learning & intelligence

- take information in
- adapt algorithms

take actions

- move gripping arm
- predict value
- generate word
- select content

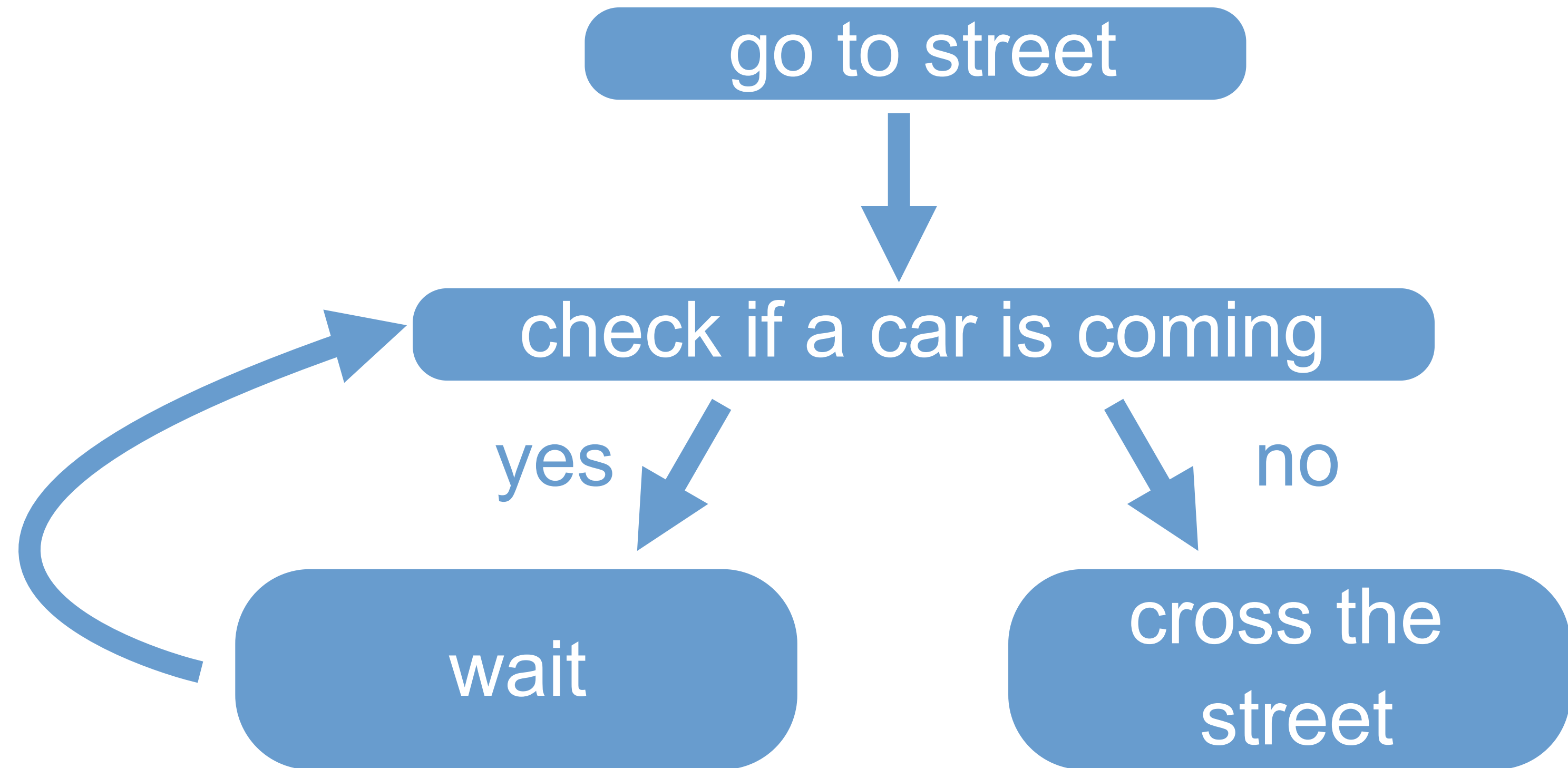
perceive environment
data as representation of
the environment

Algorithm

A clear set of instructions for solving a problem.

Composed of a finite number of well-defined steps.

... to cross a street



Adapted from Studyflix "[Was ist ein Algorithmus?](#)"

Program: Implementation of an algorithm in executable code.

Recognizing Numbers

Problem

Many handwritten
numbers that need to be
digitized.

Humans can recognize the numbers very well, but that **takes time**. Can this not be **automated**?

economic aspects of AI

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Source: Ilmer & Partner, [Informationen zum Steuerrecht](#).

Recognizing Numbers

We want to **develop a program** that takes **scans of payment slips** as input and **outputs (digital) numbers** that with **high probability** correspond to the handwritten numbers.

1. **Use domain knowledge:** we know where numbers appear on the scan and how many numbers should be there.
2. **Limit the problem:** we only need to develop an algorithm that recognizes a single number in an image.



Adapted from: Ilmer & Partner, [Informationen zum Steuerrecht](#).

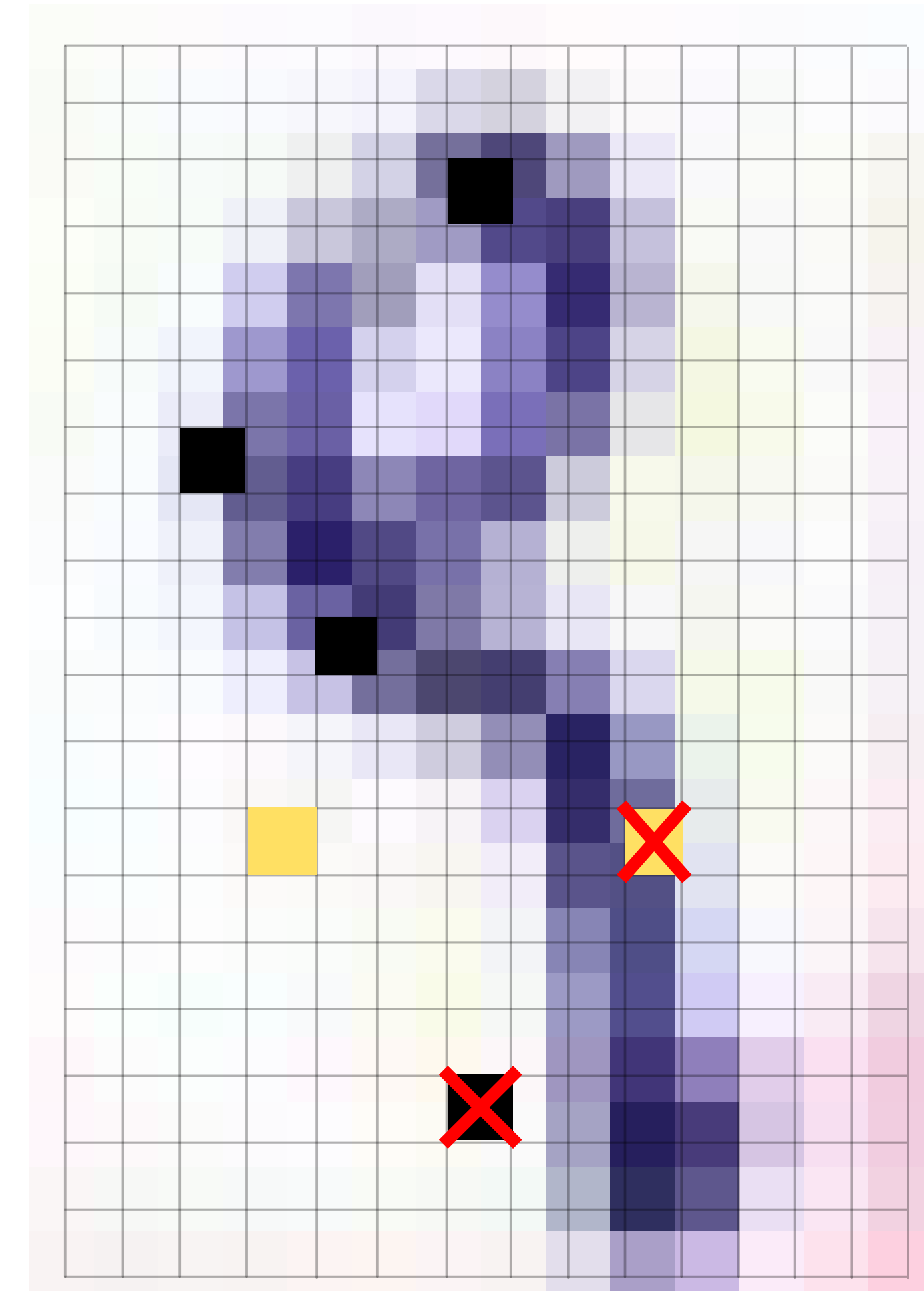
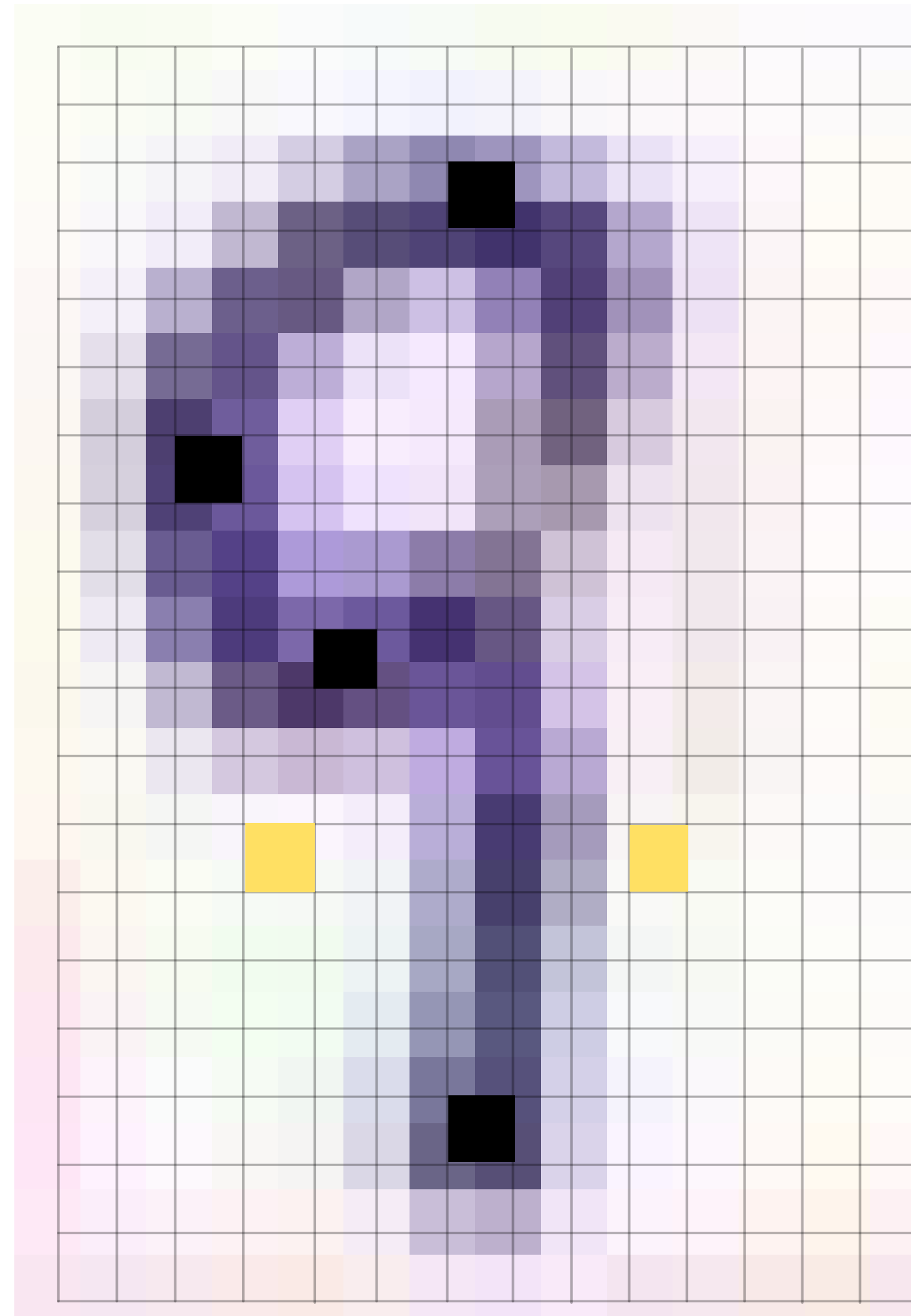
Computational thinking: *Deconstructing a problem such that it can be solved with an algorithm.*

Recognizing a Single Number

A simple idea:

- Pictures consist of pixels.
- If some of the pixels are black or white, the image must show a "Nine".

Handwriting can be very different, however.

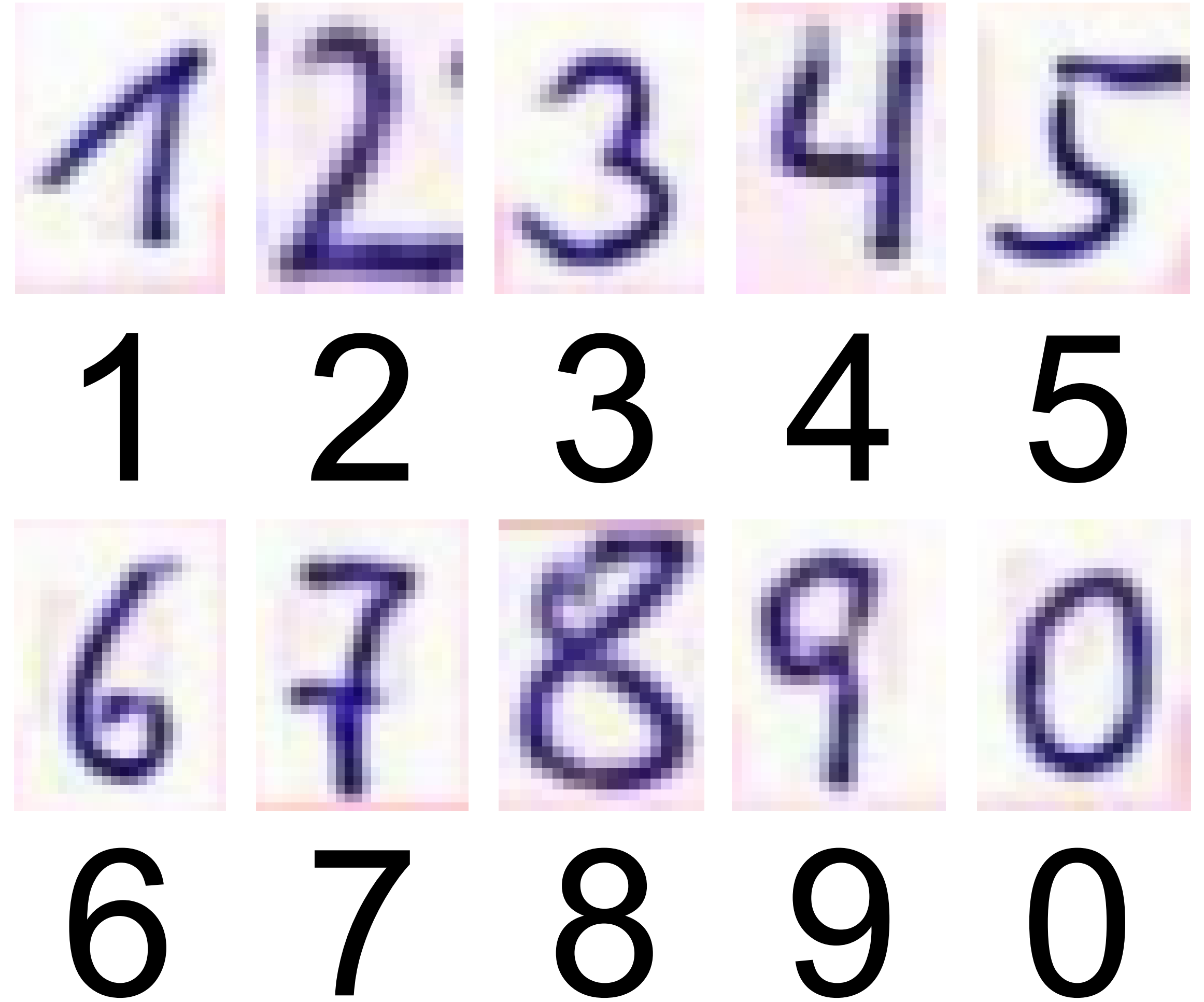


Adapted from: Ilmer & Partner, [Informationen zum Steuerrecht](#).

Machine Learning

Take information in to adapt the algorithm.

1. Enough labeled examples of numbers.
2. An algorithm that can learn from the examples.



Adapted from: Ilmer & Partner, [Informationen zum Steuerrecht](#).

Machine Learning

- Machine learning is a subfield of Artificial Intelligence.
- AI systems use machine learning to extract patterns from data, perceive their environment, make decisions, or adapt.
- **Open questions**
 - What types of machine learning algorithms are there?
 - How do they learn?

Summary

- Artificial Intelligence has many definitions.
- At its core machine learning: Algorithms that can learn from data to solve problems.
- Example: Recognizing numbers on a payment slip.
- Example for
 - **Supervised learning:** *Learning from examples.*
 - **Classification:** *Recognizing class membership of examples.*
 - **Computer Vision:** *Application area for machine learning.*